APPLYING FOR: SECURITY ENGINEER, OFFENSIVE SECURITY — ANTHROPIC

Michael Mancuso

Offensive Security Engineer | Red & Purple Team Operator

Austin, TX 512-948-2315 mancusomjm@gmail.com https://premonitionsec.com/resume
https://github.com/MichaelMancuso/MichaelMancuso/

■ ACTIVE TOP SECRET / SCI

SECURITY ENGINEER CANDIDATE

SUMMARY

Offensive security professional with **7+ years conducting red and purple team engagements** simulating advanced persistent threat (APT) actors across **cloud infrastructure (AWS, Azure), on-premises enterprise networks, and bare metal deployments**. Proven expertise in penetration testing high-value targets and translating complex attack narratives into **executive-ready risk insights that drive security roadmaps**. Skilled at developing **AI-assisted security testing tooling and custom offensive automation** that surfaces vulnerabilities commodity scanners miss. Deep cross-functional experience collaborating with engineering and detection teams on AI-specific and cloud-native attack scenarios. **Active TS/SCI clearance.**

WORK EXPERIENCE

Senior Offensive Security Engineer — *Inovalon Healthcare Data Provider*

Nov 2022 – Present

- Led **red and purple team engagements simulating advanced threat actors** against high-value healthcare data infrastructure spanning AWS and Azure cloud environments, containerized microservices, and serverless deployments — consistently revealing attack paths undiscovered by automated tooling.
- Penetration tested sensitive, high-value platforms under PCI, HIPAA, and SOX requirements; translated exploitation chains into **prioritized risk narratives for technical teams and executive leadership**, directly informing remediation roadmaps and budget decisions.
- Developed and maintained **custom offensive tooling and AI-assisted security testing scripts** (GitHub) to identify logic flaws that commodity scanners miss — including payloads for cloud IAM abuse, IDOR chains, and authentication bypass scenarios.
- **Contributed to AI-assisted security testing workflows** by integrating automated reconnaissance, exploit chaining, and finding triage into CI/CD DevSecOps pipelines using SAST and DAST toolsets — enabling continuous offensive validation as code ships.
- Executed **cross-functional purple team exercises** with engineering and detection teams, testing AI-specific and cloud-native attack scenarios and validating detection coverage across production environments.

- Conducted internal Active Directory red team assessments leveraging **Responder, Mimikatz, BloodHound, and ADRecon** to simulate credential harvesting, lateral movement, and privilege escalation across enterprise Windows domains.
- Assessed cloud-native attack surface across AWS and Azure with a holistic **IAM security perspective** — identifying over-permissioned roles, metadata service exploitation paths, and container escape vectors in Kubernetes and ECS environments.

Red Team Operator / Penetration Tester II — *Texas Comptroller of Public Accounts*

Nov 2019 – Oct 2022

- Conducted **red team engagements replicating advanced remote threat actors** across hybrid environments — from on-premises Windows Server and **OpenStack bare metal deployments** to AWS cloud infrastructure — assessing the full kill chain from initial access through impact.
- **Built and launched a new red team program from the ground up**: established rules of engagement, scoping guidelines, approved offensive methodology, charter documentation, and executive reporting cadence.
- Deployed a scalable red team operations platform within AWS featuring **custom AMIs, rolling Kali Linux upgrades, and BurpSuite Pro** — enabling concurrent multi-operator engagements against isolated target environments.
- Led **quarterly Active Directory red team evaluations**, discovering privilege escalation paths using BloodHound, ADRecon, Purple Knight, and Responder; delivered findings as board-level risk briefings.
- Produced technical and executive-level reports that **translated attack narratives into actionable risk insights**, driving leadership to fund targeted remediation and long-term architectural improvements.

Penetration Tester — *Delta Risk LLC*

Feb 2016 – Nov 2019

- Executed white-box and black-box penetration tests against client cloud and on-premises infrastructure, **simulating advanced threat actors using C2 frameworks** including Cobalt Strike, PowerShell Empire, and Starkiller.
- Developed a personal **OSINT and adversary reconnaissance methodology** using AMASS, Certificate Transparency logs, Recon-NG, and Discover — enabling tailored pre-engagement attack narratives.
- Delivered findings to **technical and executive audiences**, providing actionable remediation recommendations aligned to FISMA and NIST 800-series standards.
- Proactively automated reporting and methodology workflows, increasing consistency and operational efficiency across multi-client engagements.

Systems Engineer — *NextGen Healthcare*

Jun 2013 – Feb 2016

- Administered and secured Windows SQL and Linux PostgreSQL servers, securing HIPAA-scoped medical data at rest and in transit — providing a deep infrastructure foundation that directly informs offensive assessments.
- Used Wireshark, Network Miner, and Security Onion to establish network baselines and anomaly detection, developing defender intuition applied directly in red team engagements.

MILITARY EXPERIENCE

Cyber Operations (3D0X2) / Cyber Surety (3D0X3) — *U.S. Air Force / Texas Air National Guard — 221st Combat Communications Squadron*

Oct 2008 – Oct 2015

Honorable Discharge | SrA / E-4 | Top Secret Clearance

- Deployed and administered **fully functional multi-classification networks (TS / Secret / Unclassified) within 48 hours of theater arrival** — under extreme pressure with zero margin for configuration error.
- Managed McAfee ePO, Linux firewalls, Blue Coat proxies, VMware vCenter/ESXi, and Cisco routing/switching with cryptographic key management — broad infrastructure knowledge foundational to offensive endpoint and network testing.

CERTIFICATIONS

- EC-Council Certified Security Analyst: Penetration Testing (ECSA)
- Certified Ethical Hacker (CEH)
- AWS Certified Solutions Architect – Associate
- Microsoft Certified: Security, Compliance & Identity (SC-900)
- Microsoft Certified: Azure Fundamentals (AZ-900)
- Certified Linux Administrator (LPIC-1)
- VMware Certified Advanced Professional 6.5 – DCV

EDUCATION

Dallas Baptist University
Bachelor of Business Administration
Dallas, TX | 2001 – 2006

CORE TECHNICAL SKILLS

C2 Frameworks	Cobalt Strike, PowerShell Empire, Starkiller
AD Attack Tools	BloodHound, Mimikatz, Responder, ADRecon, Purple Knight
Web Testing	BurpSuite Pro, Metasploit, Nmap, OWASP
Cloud Offensive	AWS / Azure IAM abuse, container escapes, serverless
Scripting	Python, Bash, PowerShell
OSINT / Recon	AMASS, Recon-NG, Cert Transparency, Discover
DevSecOps	SAST, DAST, CI/CD pipeline integration